When Tooth-Numbering Detectors Fail Together: A Position-Only Baseline and Closed Gaps on Panoramic X-rays

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Abstract. Tooth-numbering detectors for panoramic X-rays reach high accuracy, but the format puts each tooth in a predictable place, so some of that accuracy could come from position alone. We measure this with a position-only baseline: a classifier that sees each tooth's bounding box and no pixels. On 425 uncropped X-rays (11,602 teeth), with cross-validation grouped by X-ray, it numbers 78.5% of teeth correctly. Three detectors (YOLOv8x, RT-DETR-l, Faster R-CNN) beat it by 14.3 to 16.7 percentage points, on all 32 tooth numbers, and this holds on two more splits. Their rare shared mistakes still follow position. On 1.4% of teeth the three detectors and the baseline are all wrong and the detectors agree on one wrong number; in 91% of these the baseline gives the same number, against under 7% by chance. These teeth are five times as likely as correct teeth to sit next to a missing tooth, and the wrong number is usually the missing tooth's number. They occur where the neighbors have closed the space. Closing a gap on purpose reproduces them: when a neighbor is moved into an erased tooth's space, all three detectors give it the missing number 21% of the time, against 0.2% when it is left open. On 634 DENTEX X-rays, detectors trained there beat position alone by 22 to 25 points, their shared errors again give the baseline's answer (85.5%), and closing a gap reproduces them 12 to 16% of the time. We release the per-tooth table and a scorer.

Keywords: Tooth numbering · Panoramic radiography · Shortcut learning · Error consistency

1 Introduction

Assigning each tooth its FDI number is the first step of most automated dental charting: a caries or a periapical lesion found by a detector is only useful once it is attached to the right tooth. Public benchmarks such as DENTEX [8] score numbering together with detection, and detectors reach high numbering accuracy (Table 1).

Panoramic X-rays are a standardized format. The patient bites on a guide, the arch runs left to right across the image, and each tooth number has a typical location. This invites a shortcut [6]: a model can get many numbers right from where a box sits, without reading the tooth. Such a shortcut would not show in

overall accuracy, and it would fail on patients who differ from the typical arch, for example after extractions.

We ask how much of tooth numbering position alone explains, and whether detectors fail where position fails. Our contributions are: (1) a position-only baseline, trained in seconds on six box features, that numbers 78.5% of teeth correctly under grouped cross-validation, with three detectors beating it by 14 to 17 points on every tooth number; (2) an error analysis showing that the errors the detectors share are position-like; (3) experiments (image shift, context masking, tooth erasure, and moving a neighbor into the space of an erased tooth) that place these errors at closed gaps, next to a missing tooth whose neighbors have moved together so that the arch looks complete; and (4) a released per-tooth benchmark with folds, reference predictions and a scorer.

2 Related Work

Tooth numbering. Detection-based numbering on panoramic X-rays is standard, with DENTEX [8] and diffusion-based detectors [9] as recent examples. DENTEX challenge entries combine segmentation with detection [10], build a YOLO-based framework [15], or use Mask R-CNN with a DiffusionDet and DINO ensemble [4]. Some methods use the arch layout as an explicit prior [1]. Evaluations report accuracy or mAP; we found none that compares a model with what position alone achieves on the same teeth.

Shortcut learning. Models can rely on cues that correlate with the label in the training data, from skin markings in dermoscopy [19] to site-specific features in chest X-rays [5] and segmentation [14]; see also [11,3]. These studies show a shortcut by removing or altering the cue. We use the same logic for a cue that is partly legitimate: position does carry information about tooth number.

Error consistency. Whether two models fail on the same inputs beyond chance is measured by error-consistency kappa [7]. We go one step further and compare the wrong answers themselves.

3 Methods

Data and splits. UFBA-425 [2] has 425 panoramic X-rays (512×512 pixels) with a mask for each tooth, labeled by FDI number (32 classes, third molars included). A tooth's box is the full extent of its mask, which gives 11,602 teeth. We use the original X-rays only. An augmented export of this data adds randomly cropped copies, which move teeth within the frame and lowered position-only accuracy by about 8 points in our earlier splits. The X-rays are split into five folds balanced by the dataset's categories. Each training fold holds out 15% of its X-rays for model selection, and every X-ray is tested once, by models that never saw it. Two further splits (other random seeds) share at most 26% and 32% of any test fold with the first.

Data use. Both datasets are public and were used as released; we collected no patient data. UFBA-425 is a set of anonymized X-rays on Figshare under CC

BY 4.0; we found no ethics statement for it. DENTEX is released by its authors under CC BY-NC-SA 4.0, and they report ethics committee approval for the X-rays from three hospitals [8].

Models. The position-only baseline is scikit-learn's gradient-boosted tree classifier (HistGradientBoosting, default settings) on six features of the labeled box: center x and y, width, height, area and aspect ratio, all relative to image size. It never sees pixels. The detectors are YOLOv8x [12], RT-DETR-l [20] and Faster R-CNN [17] with a ResNet-50 FPN backbone [13], all starting from COCO weights. They are a single-stage CNN, a transformer and a two-stage CNN, from two separate code bases. Each is trained for 30 epochs with horizontal flips off (a flip would swap left and right tooth numbers). YOLOv8x and RT-DETR-l read 640-pixel input; Faster R-CNN uses torchvision's default resizing (short side 800 pixels) and SGD with a learning rate scaled to its batch of 2. We keep the epoch with the best validation fitness ($0.1\,\mathrm{mAP}_{50} + 0.9\,\mathrm{mAP}_{50:95}$).

Scoring. Detections with confidence of at least 0.5 are matched one-to-one to labeled teeth at IoU of at least 0.5, highest confidence first, whatever number they give. A tooth with no match counts as wrong. Top-1 is the share of labeled teeth that are matched and given the right number; this puts the detectors and the baseline, which always answers, on the same teeth. We also report numbered F1, which also penalizes extra boxes. AP50 and AP50:95 are in the code release; top-1 stays the headline because the baseline has no scores. CIs are 95% intervals from 10,000 bootstrap draws of X-rays, paired when two models are compared. The decision rule for each experiment was written and frozen before it was run.

A tooth is a *joint failure* when the position-only model and all three detectors are wrong, no detector missed the tooth, and the three detectors give the same number. The *position match* is the share of joint failures where the position-only model gives that same number. Its chance level comes from shuffling the position-only answers among the joint failures.

Interventions. *Shift:* the whole image is moved sideways by 10% of its width (about two tooth widths), and the labels move with it. *Context masking:* everything outside a tooth's box, widened by k box widths and heights ($k = 0.5, 1, 2$), is blacked out. *Missing neighbor:* the next position in the arch has no labeled tooth. *Tooth erasure:* for up to three teeth per X-ray with both neighbors present, the tooth's mask (grown by 3 px) is filled in by inpainting [18], and we ask whether a neighbor that was numbered right now takes the erased tooth's number. Controls are the teeth two places away and a sham inpaint over bone. *Space closure:* for each missing tooth with both neighbors present, S is the shortest distance between the neighbors' masks divided by the missing tooth's median width ($S = 0$ when they touch). *Gap closure:* after erasing a tooth, its distal neighbor (the next one toward the back of the mouth) is cut out and pasted back shifted toward the gap, by half or all of the distance at which it touches the tooth on the other side, with its box moved to match. In a variant, *tipping*, the neighbor is instead rotated about its root apex until its crown touches.

Table 1. Top-1 on 11,602 teeth (first split, 95% CI) and each detector's gap over the position-only model. F1 also penalizes extra boxes. The last column gives the gap on the two further splits.

	Top-1 (%)	Gap (points)	Numbered F1	Gap, other splits
Position-only	78.5 (77.0 to 80.0)			
YOLOv8x	95.0 (94.2 to 95.8)	16.5 (15.1 to 17.9)	95.0	15.8, 16.4
RT-DETR-l	95.2 (94.4 to 95.9)	16.7 (15.3 to 18.1)	92.3	15.9, 16.4
Faster R-CNN	92.8 (91.9 to 93.7)	14.3 (13.0 to 15.7)	89.6	13.8, 14.3

4 Results

Position explains most; detectors add 14 to 17 points. Position alone numbers more than three in four teeth (Table 1). Each detector beats it on all 32 tooth numbers. The gap is smallest at the third molars (4.3 to 5.8 points) and largest at the lower central incisors (up to 25.5). If the baseline is given the detectors' own boxes instead of the labeled ones, it does slightly better (79.0 to 80.2%) and the gap is 14.5 to 16.2 points, so the labeled boxes do not inflate its score. On the two further splits, every gap falls inside the first split's CI. YOLOv8x and RT-DETR-l tie on top-1 (difference 0.17 points, where the smallest difference we could detect is 0.50), but RT-DETR-l leaves about 1.9 unmatched boxes per X-ray against 0.4 for YOLOv8x, which lowers its F1 by 2.7 points.

Shared errors follow position. 161 teeth (1.4%) are joint failures. In 91.3% of them the position-only model gives the same wrong number as the detectors; shuffling its answers gives at most 6.8% (95th percentile). In 94.4% the wrong number is the next tooth over in the same quadrant. The detectors are about as confident on these teeth as on correct ones (AUROC 0.53 to 0.66, where 0.5 is chance), so a confidence cutoff does not find them, and neither does a rule based on the detectors' second choices. About 10% of joint failures may be label errors [16]; without them the position match is 91.7%.

Not position in the frame, not distant teeth. Shifting the image by two tooth widths leaves the position-only model right on under 9% of teeth, but changes the detectors' top-1 by under 0.3 points. Blacking out everything beyond one box width costs Faster R-CNN and RT-DETR-l about 26 points (YOLOv8x 9.6), counting teeth that are no longer detected as wrong. Even so, among the joint failures that are still detected, 74 to 93% keep the shared wrong number.

Next to missing teeth. 44.7% of joint failures have a missing neighbor, against 8.5% of teeth that all three detectors get right (ratio 5.3, CI 4.2 to 6.6). In 67 of the 72 joint failures next to a gap (93%), the wrong number is the missing tooth's. Teeth that only the position-only model gets wrong have a missing neighbor 10.5% of the time, so gaps are not where position fails in general, only where position and the detectors fail together.

At closed gaps. Across 372 missing teeth with both neighbors present, the space left between the neighbors separates gaps with a joint failure from gaps

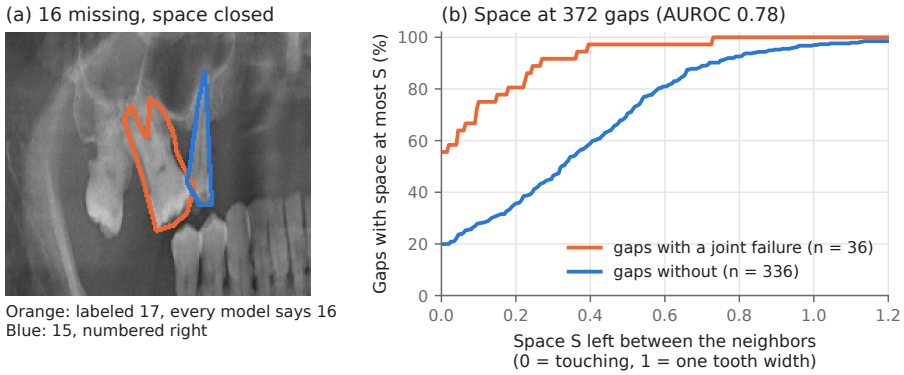


Fig. 1. (a) A closed gap: tooth 16 is missing and its neighbors' outlines touch. All three detectors and the position-only model number tooth 17 as 16. (b) Share of the 372 gaps with both neighbors present whose space is at most S, for gaps with and without a joint failure.

without one (Fig. 1; AUROC 0.78, CI 0.70 to 0.84; median S 0.00 against 0.32). 56% of these errors sit at gaps whose neighbors touch, against 20% of other gaps. Each detector's own errors follow closure about as well (AUROC 0.74 to 0.79), and better than the position-only model's (0.68).

An open gap is not enough. After a tooth is erased, a correctly numbered neighbor takes its number in 2.0% (YOLOv8x), 4.4% (RT-DETR-l) and 5.7% (Faster R-CNN) of cases, against 0 to 0.2% for both controls. But all three detectors do it to the same neighbor in only 4 of 2,260 cases (0.18%).

Closing the gap reproduces the shared error. For the same erased teeth, we moved the distal neighbor sideways until it touched the tooth on the other side. Of 1,114 moved teeth that all three detectors numbered right on the intact X-ray, all three gave the missing tooth's number for 0.18% with the space open, 2.9% when moved halfway and 20.8% when moved all the way (20.6 points above open, CI 18.2 to 23.1). Each detector alone did so 38 to 42% of the time. Models trained on the two further splits give the same result (22.5% and 21.8% at full closure, at most 0.1% open). Tipping the neighbor, which is closer to how real teeth drift, gives 9.4% (CI 7.7 to 11.0; 9.3% on the second split's models). At real gaps whose neighbors almost touch ($S \leq 0.1$), 22% hold a joint failure, about 11% per neighbor. Cutting the tooth out and pasting it back unmoved gives 0.0%, so the pasted edges do not cause the effect. In these cases the moved tooth looks the same as before, yet the detectors number it by the slot it now occupies.

A second dataset. We repeated the main steps on 634 DENTEX X-rays (18,095 teeth) from other clinics [8]. Position alone numbers 66.9% of these teeth. The UFBA-trained detectors, run unchanged, lose accuracy under this shift (YOLOv8x misses many teeth; Faster R-CNN is often one position off), though their shared errors still follow position (87.5%). We therefore also trained

all three on DENTEX with five-fold cross-validation. They then beat position alone by 22 to 25 points. Their 303 shared errors give the baseline's answer 85.5% of the time (UFBA: 91.3%) and sit next to a missing tooth 3.5 times as often as correct teeth (UFBA: 5.3). Both fall short of the targets we set in advance (85.7% and 4.2), so we call this a partial replication. Two more training runs on the same folds give the same picture: margins of 24 to 26 points, a position match of 85.5% and 84.2%, and a gap ratio of 3.3 and 3.2; 213 of the 410 teeth that are a shared error in any run are one in all three. Part of the weaker gap link depends on which X-rays are in the data: after weighting DENTEX to UFBA's mix of missing teeth per X-ray, the ratio is 3.9. The interventions also work on DENTEX, with smaller effects (tables in the code release). At each X-ray's stored size, sliding a neighbor into the erased tooth's space gives the missing number 15.7% of the time (0.4% open) and tipping it 7.0% (0.4% open), about three quarters of the UFBA rates (one training run). On X-rays shrunk to 640 pixels wide the rates are lower (slide 11.8%, 12.0% and 12.6%, tip 4.8%, 5.5% and 6.5% on three training runs; our target of 5% for the tip was met on two of the three), mostly because Faster R-CNN reads the shrunken X-rays less well. The individual teeth that fill differ between runs: of 341 slide fills in any of the three runs, 57 occur in all three.

5 Discussion

The detectors read teeth: shifting the image barely changes their answers, and they beat the position-only model on every tooth number. Yet their rare shared errors are the position-only model's errors, and our results suggest where they come from. When a tooth is missing and its neighbors have moved together, the arch shows no gap. Counting along it gives the tooth next to the closed space the missing tooth's number, and that tooth also sits roughly where the missing one would. Appearance, context and position then point the same way, and every model follows. An erased tooth leaves a full-width empty space, which does not create this agreement; closing the space does.

Two practical points follow. First, a position-only baseline is cheap and should be reported with tooth-numbering results: on standardized panoramic X-rays it sets a high floor, and the margin above it shows what a model adds. Second, confidence does not find the errors that models share. A check from outside the model works better: look at the numbering next to every missing tooth, especially where the space has closed. In our data this check covers 97% of the shared errors at gaps, but it also flags many gaps without one (precision 13%), because most gaps are partly closed. A version that uses only a model's own output flags teeth next to a position the model left empty, and teeth that share a number. It flags 12 to 16% of teeth and catches 41 to 47% of the shared errors on every split, against 17 to 32% for the same number of least-confident teeth. We had set a target of at least 1.5 times as many as confidence, which it reached on two of three splits. Training with simulated missing and closed teeth did not

remove the error: YOLOv8x numbered teeth next to real gaps no better than a retrained control (+0.1 points, CI −1.6 to 1.7).

Limitations. UFBA is one dataset from one source, with a single annotation per tooth and no patient data. The DENTEX replication is partial, and its stored-size intervention results rest on one training run. The interventions move or tip a tooth within the image plane only, but real teeth also rotate, and outlines that touch in a flat image may not touch in the mouth. Third molars and longer gaps, which hold most missing positions, are outside the closure analysis. The rates are stable across splits and training runs, but the individual teeth are not: about a third of the joint failures change when the same model is retrained (overlap 0.63, against 0.54 to 0.57 across splits). About 10% of joint failures may be label errors.

6 Conclusion

On uncropped panoramic X-rays, position alone numbers 78.5% of teeth and detectors add 14 to 17 points. The few errors they share follow position and concentrate next to missing teeth whose neighbors have closed the space. On a second dataset, DENTEX, the same pattern holds with a weaker link to gaps and smaller intervention effects. We release the per-tooth table and scorer so that new models can be measured against the baseline and on these shared failures.

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